

Python Notes: Data Types in Python (Complete)

1. Numeric Data Types

Numeric data types store numerical values.

Integer (int): Whole numbers without decimals.

Example: num=100

Float (float): Numbers with decimal values.

Example: price=99.99

Complex (complex): Numbers with real and imaginary parts.

Example: x=2+3j

2. String Data Type (str)

Strings store text data.

Example:

name='Python'

message='Hello'

Strings can use single, double, or triple quotes.

3. Boolean Data Type (bool)

Boolean stores only two values:

True

False

Example:

is_student=True

4. List Data Type (list)

Lists store multiple items.

Example:

fruits=['Apple','Banana','Mango']

Characteristics:

Ordered

Mutable

Allows duplicates

5. Tuple Data Type (tuple)

Tuples store multiple items that cannot be modified.

Example:

```
numbers=(10,20,30)
```

Characteristics:

Ordered

Immutable

6. Set Data Type (set)

Sets store unique values.

Example:

```
colors={'Red','Green','Blue'}
```

Characteristics:

Unordered

No duplicates

7. Frozen Set Data Type (frozenset)

Frozen set is an immutable version of a set.

Example:

```
colors=frozenset(['Red','Green','Blue'])
```

Characteristics:

Unordered

Immutable

No duplicate values

8. Dictionary Data Type (dict)

Dictionary stores data as key-value pairs.

Example:

```
student={'name':'John','age':20}
```

Characteristics:

Mutable

Keys must be unique

Type Function

Use `type()` to identify a data type.

Example:

```
print(type(10))
```

Summary

Python data types define the type of values a variable refers to. Common types include numeric, string, boolean, list, tuple, set, frozenset, and dictionary.